

Task 1 — Create Users

Create Users with Home Directories

None

```
sudo useradd -m tokyo
sudo useradd -m berlin
sudo useradd -m professor
```

What `-m` does

Creates:

None

```
/home/username
```

Set Passwords

None

```
sudo passwd tokyo
sudo passwd berlin
sudo passwd professor
```

It will ask:

None

```
New password:
Retype new password:
```

Verify Users

Check `/etc/passwd`

None

```
cat /etc/passwd | grep tokyo  
cat /etc/passwd | grep berlin  
cat /etc/passwd | grep professor
```

Expected output example:

None

```
tokyo:x:1001:1001:./home/tokyo:/bin/sh
```

Verify Home Directories

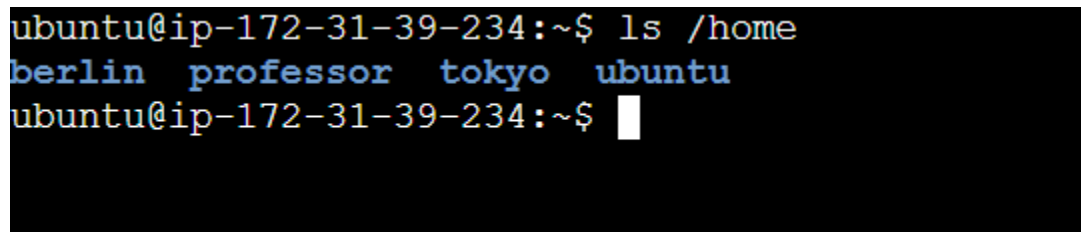
None

```
ls /home
```

Expected:

None

```
berlin professor tokyo
```



```
ubuntu@ip-172-31-39-234:~$ ls /home  
berlin professor tokyo ubuntu  
ubuntu@ip-172-31-39-234:~$
```

Screenshot to Take

Take screenshots of:

- `ls /home`

- `/etc/passwd` output

```
dnsmasq:x:999:65534:dnsmasq:/var/lib/misc:/usr/sbin/nologin
tokyo:x:1001:1001::/home/tokyo:/bin/sh
berlin:x:1002:1002::/home/berlin:/bin/sh
professor:x:1003:1003::/home/professor:/bin/sh
ubuntu@ip-172-31-39-234:~$
```

Task 2 — Create Groups

Create Groups

None

```
sudo groupadd developers
sudo groupadd admins
```

Verify Groups

None

```
cat /etc/group | grep developers
cat /etc/group | grep admins
```

Expected:

None

```
developers:x:1004:
admins:x:1005:
```

```
ubuntu:x:1000:
docker:x:109:
tokyo:x:1001:
berlin:x:1002:
professor:x:1003:
developers:x:1004:
admins:x:1005:
```

None

```
cat /etc/group | grep developers
```

```
ubuntu@ip-172-31-39-234:~$ cat /etc/group | grep developers
developers:x:1004:
ubuntu@ip-172-31-39-234:~$
```

Task 3 — Assign Users to Groups

Add Users to Groups

tokyo → **developers**

None

```
sudo usermod -aG developers tokyo
```

berlin → **developers + admins**

None

```
sudo usermod -aG developers berlin
sudo usermod -aG admins berlin
```

professor → **admins**

None

```
sudo usermod -aG admins professor
```

Verify Group Membership

None

```
groups tokyo  
groups berlin  
groups professor
```

Expected:

None

```
tokyo : tokyo developers  
  
berlin : berlin developers admins  
  
professor : professor admins
```

```
ubuntu@ip-172-31-39-234:~$ groups professor  
professor : professor admins  
ubuntu@ip-172-31-39-234:~$ groups tokyo  
tokyo : tokyo developers  
ubuntu@ip-172-31-39-234:~$ groups berlin  
berlin : berlin developers admins  
ubuntu@ip-172-31-39-234:~$ groups professor  
professor : professor admins  
ubuntu@ip-172-31-39-234:~$
```

None

```
groups berlin
```

Task 4 — Shared Directory

Create Directory

None

```
sudo mkdir /opt/dev-project
```

Change Group Owner

None

```
sudo chgrp developers /opt/dev-project
```

Set Permissions

None

```
sudo chmod 775 /opt/dev-project
```

Meaning of 775

775775775

Actually means:

None

```
Owner   = rwx = 7
```

```
Group   = rwx = 7
```

```
Others  = r-x = 5
```

Verify Permissions

None

```
ls -ld /opt/dev-project
```

Expected:

None

```
drwxrwxr-x
```

```
ubuntu@ip-172-31-39-234:~$ sudo mkdir /opt/dev-project
ubuntu@ip-172-31-39-234:~$ sudo chgrp developers /opt/dev-project
ubuntu@ip-172-31-39-234:~$ sudo chmod 775 /opt/dev-project
ubuntu@ip-172-31-39-234:~$ ls -ld /opt/dev-project
drwxrwxr-x 2 root developers 4096 May 23 14:06 /opt/dev-project
ubuntu@ip-172-31-39-234:~$
```

Test File Creation

Create file as tokyo

None

```
sudo -u tokyo touch /opt/dev-project/tokyo.txt
```

Create file as berlin

None

```
sudo -u berlin touch /opt/dev-project/berlin.txt
```

Verify

None

```
ls -l /opt/dev-project
```

Expected:

None

```
tokyo.txt
```

```
berlin.txt
```

```
ubuntu@ip-172-31-39-234:~$ ls -l /opt/dev-project
total 0
-rw-r--r-- 1 berlin berlin 0 May 23 14:10 berlin.txt
-rw-r--r-- 1 tokyo tokyo 0 May 23 14:10 tokyo.txt
ubuntu@ip-172-31-39-234:~$
```

```
ubuntu@ip-172-31-39-234:~$ ls -ld /opt/dev-project
drwxrwxr-x 2 root developers 4096 May 23 14:10 /opt/dev-project
ubuntu@ip-172-31-39-234:~$
```

Take screenshots of:

- `ls -ld /opt/dev-project`
- `ls -l /opt/dev-project`

Task 5 — Team Workspace

Create User

None

```
sudo useradd -m nairobi
```

Set password:

None

```
sudo passwd nairobi
```

Create Group

None

```
sudo groupadd project-team
```

Add Users to Group

None

```
sudo usermod -aG project-team nairobi  
sudo usermod -aG project-team tokyo
```

Create Directory

None

```
sudo mkdir /opt/team-workspace
```

Set Group Ownership

None

```
sudo chgrp project-team /opt/team-workspace
```

Set Permissions

None

```
sudo chmod 775 /opt/team-workspace
```

Test File Creation

None

```
sudo -u nairobi touch /opt/team-workspace/test.txt
```

Verify

None

```
ls -ld /opt/team-workspace
```

```
ls -l /opt/team-workspace
```

Final Markdown Documentation

Paste this into `day-09-user-management.md`

None

```
# Day 09 Challenge
```

```
## Users & Groups Created
```

```
### Users
```

- tokyo
- berlin
- professor
- nairobi

```
### Groups
```

- developers
- admins
- project-team

```
---
```

```
## Group Assignments
```

```
| User | Groups |
|-----|-----|
| tokyo | developers, project-team |
| berlin | developers, admins |
| professor | admins |
| nairobi | project-team |
```

```
---
```

Directories Created

Directory	Group	Permissions
/opt/dev-project	developers	775
/opt/team-workspace	project-team	775

Commands Used

```
```bash
sudo useradd -m tokyo
sudo useradd -m berlin
sudo useradd -m professor
sudo passwd tokyo
sudo passwd berlin
sudo passwd professor

sudo groupadd developers
sudo groupadd admins

sudo usermod -aG developers tokyo
sudo usermod -aG developers berlin
sudo usermod -aG admins berlin
sudo usermod -aG admins professor

groups tokyo
groups berlin
groups professor

sudo mkdir /opt/dev-project
sudo chgrp developers /opt/dev-project
sudo chmod 775 /opt/dev-project
```

```
sudo -u tokyo touch /opt/dev-project/tokyo.txt
sudo -u berlin touch /opt/dev-project/berlin.txt

sudo useradd -m nairobi
sudo passwd nairobi

sudo groupadd project-team

sudo usermod -aG project-team nairobi
sudo usermod -aG project-team tokyo

sudo mkdir /opt/team-workspace
sudo chgrp project-team /opt/team-workspace
sudo chmod 775 /opt/team-workspace

sudo -u nairobi touch /opt/team-workspace/test.txt
...

What I Learned

1. How to create Linux users and groups
2. How Linux permissions work using chmod
3. How shared group access works in Linux
```

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## Useful Troubleshooting Commands

### Check User Exists

None

```
id tokyo
```

---

## Check Group Exists

None

```
getent group developers
```

---

## Check Permissions

None

```
ls -ld /opt/dev-project
```

---

## Switch User

None

```
su - tokyo
```

Exit back:

None

```
exit
```

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## Important DevOps Concepts Learned

- Linux user management
- Group-based access control
- Shared team directories
- File permissions
- Multi-user collaboration
- Security basics in Linux